

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0704

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

ANSWERS

1. Compare Star Topology and Bus Topology in terms of Cost, Reliability and Performance

Answer:

Star topology and **Bus topology** differ mainly in their cost, reliability, and network performance.

Cost:

- **Star topology** is generally more expensive because it requires a **central device such as a switch or hub**.
- Each node requires a separate cable connection to the central device.
- **Bus topology** is cheaper because all devices share a **single backbone cable**.
- Therefore, bus topology is easier and less expensive to install.

Reliability:

- Star topology is more reliable because the failure of one node or its cable does not affect the other nodes.
- However, if the central switch or hub fails, the entire network can stop functioning.
- In bus topology, failure of the main backbone cable can bring down the entire network.

Performance:

- Star topology generally provides better performance because communication is controlled through the central switch.
- It reduces collisions and manages data transmission efficiently.
- In bus topology, all devices share the same communication medium.
- As more devices are added, network traffic increases and performance decreases.

Comparison Table:

Factor	Star Topology	Bus Topology
Cost	Higher	Lower
Cabling	More	Less
Reliability	High	Lower
Performance	Better	Decreases with traffic
Main failure point	Switch/Hub	Backbone cable

Conclusion:

Star topology is preferred for reliable and high-performance networks, while **bus topology is suitable for small networks where low cost and simple installation are important**. The source specifically notes that star has better reliability and performance, while bus has lower installation cost.

2. Compare Mesh Topology and Star Topology in terms of Fault Tolerance

Answer:

Mesh topology provides higher fault tolerance than **star topology** because of its multiple communication paths.

Mesh Topology:

- In mesh topology, each node is connected to multiple other nodes.
- Therefore, there are **multiple paths** available for transmitting data.
- If one link fails, data can travel through another available path.
- This prevents a single link failure from disrupting communication.
- Hence, mesh topology provides **very high fault tolerance**.

Star Topology:

- In star topology, each node is connected to a central switch or hub.
- Failure of an individual node normally does not affect the rest of the network.
- However, failure of the central switch or hub can cause the **entire network to stop functioning**.
- Therefore, its fault tolerance is considered moderate.

Comparison:

Factor	Mesh	Star
Communication paths	Multiple	Mainly through central device
Fault tolerance	Very high	Moderate
Individual link failure	Network can continue	Usually affects only that node
Central device failure	No single central device	Can stop entire network
Reliability	Very high	High

Example:

A mesh network can be used in **critical environments** where continuous communication is important because an alternate path can be used when one connection fails.

Conclusion;

Mesh topology is more fault-tolerant and robust than star topology because it provides alternate paths for communication.

3. Compare Bus Topology and Mesh Topology in terms of Installation Complexity

Answer:

Bus topology is much simpler to install than **Mesh topology**.

Bus Topology:

- Bus topology uses a **single backbone cable**.
- Devices are connected to this common cable.
- It requires fewer cables and connections.
- Installation is simple and straightforward.
- Therefore, it is suitable for small networks.

Mesh Topology:

- In mesh topology, nodes are connected to multiple or all other nodes.
- This requires a large number of cables and ports.
- As the number of devices increases, the number of connections increases significantly.
- Installation becomes complicated, time-consuming, and expensive.

Comparison Table:

Factor	Bus Topology	Mesh Topology
Main structure	Single backbone	Multiple interconnections
Installation	Easy	Complex
Cabling	Less	Large amount
Number of connections	Few	Very high
Cost	Low	High
Scalability of installation	Simple for small networks	Becomes difficult as nodes increase

Example:

For a small laboratory with a few computers, bus topology can be easier to install. A large critical network requiring high redundancy may use mesh topology despite its installation complexity.

Conclusion:

Bus topology is easier, faster, and cheaper to install, whereas **mesh topology provides greater reliability but requires complex installation and more resources**.

4. Compare Hybrid Topology and Star Topology in terms of Scalability

Answer:

Hybrid topology provides greater scalability than star topology because it combines different network topologies and allows flexible expansion.

Hybrid Topology:

- Hybrid topology combines **two or more different topologies**, such as star, mesh, or bus.
- Different network sections can be designed according to organizational requirements.
- New network segments can be added without significantly affecting the existing structure.
- This makes hybrid topology highly flexible and scalable.

Star Topology:

- Star topology can also be expanded by adding more devices.
- However, all devices depend on the central switch or hub.
- The number of available ports limits the number of devices that can be connected.
- Once the central device reaches its port capacity, additional hardware is required.

Comparison Table:

Factor	Hybrid Topology	Star Topology
Scalability	Very high	Moderate
Expansion	Flexible	Limited by switch/hub ports
Structure	Combination of topologies	Centralized
Flexibility	Very high	Moderate
Additional hardware	Can be added according to need	Required after port limit
Suitable for	Large organizations	Small/medium networks

Example:

A large organization can use a hybrid topology by using **star topology in individual departments and connecting those departments using another topology**. This allows the network to grow according to organizational needs.

Conclusion:

Hybrid topology is more scalable than star topology because it allows different topologies and network segments to be combined and expanded flexibly.

5. Compare Star, Mesh, Bus and Hybrid Topologies in terms of Maintenance

Answer:

The four topologies have different maintenance requirements based on their structure and complexity.

Star Topology:

- Star topology is relatively easy to maintain.
- Problems can be quickly identified and isolated to a particular node or cable.
- Troubleshooting is therefore easier.
- However, the central switch or hub requires proper maintenance.

Mesh Topology:

- Mesh topology is highly reliable but difficult to maintain.
- It contains a large number of connections.
- Troubleshooting individual connections can be time-consuming.
- Skilled administrators may be required.

Bus Topology:

- Bus topology is more difficult to maintain than star topology.
- A fault in the backbone cable can affect the entire network.
- Finding the exact location of a backbone fault can be difficult.

Hybrid Topology:

- Hybrid topology has **moderate maintenance complexity**.
- It provides flexibility but combines multiple topologies.
- Administrators need to monitor different network sections carefully.
- Skilled administration may be required.

Comparison Table:

Topology	Maintenance	Main Reason
Star	Easy	Faults can be isolated easily
Mesh	Difficult	Large number of connections
Bus	Difficult	Backbone failure can affect entire network
Hybrid	Moderate	Multiple topologies require monitoring

Overall Comparison:

Star → Easiest maintenance

Hybrid → Moderate maintenance

Bus → Difficult maintenance

Mesh → Most complex maintenance

Conclusion:

Star topology is the easiest to maintain because faults can be isolated to individual nodes or cables.

Mesh topology is the most difficult to maintain because of its large number of connections. Hybrid topology requires careful monitoring, while bus topology makes backbone faults difficult to locate.